

CHAPTER 3

Hyperbolic Systems—general facts

This chapter is design to give an idea of the general results of hyperbolic theory. Since such a theory covers a rather vast landscape and it contains very technical results our exposition is bound to be quite sketchy. Nevertheless, in the following chapters we will prove all the results we need in this book for the special setting of area preserving two dimensional maps (see section 4.4 and Problem 3.7 for Oseledets and chapter 7 for the results on foliations).

3.1 Hyperbolicity

Our goal in this section is to introduce and discuss a class of systems for which we can hope to extend the results of the previous section. As we have seen, the key property the we used in the study of the Arnold cat were the expanding and contracting properties of the map. These are generalized in the following definition.

Definition 3.1.1 *By Hyperbolic System (with discrete time) we mean a Dynamical System (X, T, μ) such that X is a smooth compact Riemannian manifold (possibly with boundary), T is almost everywhere differentiable and there exists two measurable families of invariant¹ subspaces $E^u(x), E^s(x) \in \mathcal{T}_x X$ transversal at almost each point,² and measurable functions $\nu(x) > 1, c(x) > 0$ such that for almost all $x \in X$*

$$\begin{aligned} \|D_x T^n v\| &\geq c(x)^{-1} \nu(x)^n \|v\| \quad \forall v \in E^u(x) \\ \|D_x T^n v\| &\leq c(x) \nu(x)^{-n} \|v\| \quad \forall v \in E^s(x). \end{aligned}$$

If the functions c, ν can be chosen constant, then the system is called Uniformly Hyperbolic. If, in addition, T is a diffeomorphism, then the system is called Anosov (or sometimes C or U system).

¹That is $D_x T E^{s(u)}(x) = E^{s(u)}(Tx)$.

²That is, $E^u(x) \cap E^s(x) = \{0\}$ and $E^u(x) \oplus E^s(x) = \mathcal{T}_x X$ a.e.

The condition in Definition 3.1.1 is essentially equivalent to saying that two very close initial conditions almost certainly will grow apart at an exponential rate. This corresponds to a strong instability with respect to the initial conditions and characterizes the sense in which the dynamics of hyperbolic systems is a very complex one. Such complex behavior has recently captured the popular fantasy under the ambiguous name of *chaos*.

3.1.1 Examples

3.1.1.a Rotations

Clearly the rotations are not hyperbolic since $DT = 1$.

3.1.1.b Dilation

One can easily see that such a system is expanding, hence $E^u = \mathbb{R}$ and $E^s = \emptyset$.

3.1.1.c Arnold cat

We have studied it in detail in the previous chapter.

3.1.1.d Baker

In this case one direction is expanding and one is contracting, $\dim E^u = \dim E^s = 1$

A more general notion of hyperbolicity is the one of *hyperbolic set*.

Definition 3.1.2 *Given a diffeomorphism T of a manifold X , we say that $\Lambda \subset X$ is hyperbolic if Λ is compact, $T(\Lambda) = \Lambda$ and there exists two measurable families of invariant subspaces $E^u(x), E^s(x) \in \mathcal{T}_x X$ transversal at each point and measurable functions $\nu(x) > 1, c(x) > 0$ such that for all $x \in \Lambda$*

$$\begin{aligned} \|D_x T^n v\| &\geq c(x)^{-1} \nu(x)^n \|v\| \quad \forall v \in E^u(x) \\ \|D_x T^n v\| &\leq c(x) \nu(x)^{-n} \|v\| \quad \forall v \in E^s(x). \end{aligned}$$

If the constants c, ν can be chosen independently of $x \in \Lambda$ then Λ is called Uniformly Hyperbolic.

✕ put def of: hyp. measure,
partial hyp., projective hyp.,
dominate split.

3.1.2 Examples

3.1.2.a Solenoid

...

3.1.2.b Smale Horseshoe

In this case the set Λ is the one constructed in Examples 1.4.1 and $\dim E^s = \dim E^u = 1$.

3.1.2.c Hyperbolic Julia sets

if possible

3.1.2.d Forced pendulum

Same situations as for the horseshoe, see Examples 1.8.1.

A first interesting result concerning hyperbolic structures is the following.

Theorem 3.1.3 *For an Anosov map the distributions $E^u(x)$, $E^s(x)$ are everywhere defined, everywhere transversal and continuous.*

PROOF. For each $x \in X$ consider a neighborhood $U(x)$ contained in a chart. Use the Cartesian structure of the chart to identify tangent vectors at different points.³ Consider $\{y_i\}$, $\lim_{i \rightarrow \infty} y_i = x$. There must exist a subsequence such that $E^u(y_i)$ converge since the set (Grassmannian) of subspaces (with the obvious topology) is compact. Let us call $E_+(x)$ one such limit. For each $v \in E_+(x)$ holds

$$\|D_x T^n v\| \geq \|D_{y_{i_k}} T^n v_k\| (1 - \varepsilon) \geq (1 - 2\varepsilon) c^{-1} \nu^n \|v\|$$

for some n_k large enough and some $v_k \in E^u(y_{i_k})$. Thus $E_+(x)$ is contained in the unstable subspace at x . In the same manner it is easy to show that if v is a vector expanding exponentially in the future, then it must belong to $E_+(x)$. The same holds for the contracting space $E_-(x)$, in addition the two subspaces are obviously transversal (see Problem 3.1). Accordingly we have a splitting defined everywhere and continuous. This in particular implies that their dimension is constant and that there must be a minimal “angle” between them. $\square$

Definition 3.1.1 it is not particularly helpful in concrete cases since, in general, it is not clear how to verify if a system is hyperbolic or not. A first step toward a better understanding is contained in the next section, then in chapter 4 we will study this issue further.

³Or, in a Riemannian manifold, one can use the Riemannian connection. Anyhow it is irrelevant: any connection will do.

3.2 Lyapunov exponents and invariant distributions

We start by a different and very helpful characterization of hyperbolicity obtained by introducing the so called Lyapunov Exponents (LE).

Definition 3.2.1 *For each $x \in X$, $v \in \mathcal{T}_x X$ we define*

$$\lambda(x, v) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \|D_x T^n v\|.$$

If $\lambda(x, v)$ exists it is called “Lyapunov exponent” (LE).

It is interesting to notice that $\lambda(Tx, D_x T v) = \lambda(x, v)$ (see Problem 3.2). Moreover it should be clear that, if the system is ergodic and the map invertible, then $\lambda(x, v)$, if it exists, can assume only finitely many values (see Problem 3.4).

The existence and properties of the LE have been studied in detail. Here we satisfy ourselves with a basic fact.

Theorem 3.2.2 (Oseledets [69], [76]) *For each (X, T, μ) , X union of compact manifolds, T almost everywhere differentiable, if*

$$\int_X \|\log D_x T\| d\mu < \infty, \quad (3.2.1)$$

then, for almost all $x \in X$, there exists numbers $\{\lambda_1, \dots, \lambda_{s(x)}\}$ and a flag of subspaces

$$\{0\} = V_0 \subset V_1(x) \subset \dots \subset V_{s(x)-1}(x) \subset V_{s(x)}(x) = \mathcal{T}_x X,$$

such that, for all $v \in V_k(x) \setminus V_{k-1}(x)$, the LE $\lambda(x, v)$ exists and equal λ_k .

Note that if T is invertible, $\{\lambda_i(x)\}$ is equal a.e. to $\{-\lambda_i^-(x)\}$ where $\{\lambda_i^-(x)\}$ are the LE of (X, T^{-1}, μ) (see Problem 3.5).

We will not prove Theorem 3.2.2 in the above generality (see ??? for a proof), but see section 4.4 for a very constructive proof in an important, but special, case.

To appreciate the relevance of the LE, in the present context, consider the following.

Theorem 3.2.3 *A system (X, T, μ) , where X is a Riemannian manifold and T is a diffeomorphism, is hyperbolic iff for almost all $x \in X$*

$$\lambda(x, v) \neq 0 \quad \forall v \in \mathcal{T}_x X, \quad v \neq 0.$$

PROOF. Keep in mind that the L.E. are defined a.e. by Theorem 3.2.2 if the system is hyperbolic, then all the LE are different from zero. The other implication is almost as trivial. Define $E^s(x) = \{v \in \mathcal{T}_x \mid \lambda(x, v) < 0\}$; then consider the Dynamical system (X, T^{-1}, μ) and its LE $\lambda^-(x, v)$ and define $E^u(x) = \{v \in \mathcal{T}_x \mid \lambda^-(x, v) < 0\}$. Next, let

$$\rho(x) = \sup\{\lambda(x, v), \lambda^-(x, w) \mid v \in E^s(x), w \in E^u(x)\}$$

clearly $\rho(x) < 0$ a.e.. Then setting $\nu(x) = e^{-\rho(x)/2}$ and

$$c(x) = \sup\{\nu(x)^n \|D_x T^n v\|, \nu(x)^n \|D_x T^{-n} w\| \mid v \in E^s(x); w \in E^u(x)\}_{n \in \mathbb{N}},$$

the theorem is proven. $\square$

3.2.1 Examples

3.2.1.a Logistic map

do by conjugation and by showing via a dyn. proof that $\lambda(x) > 0$ for each $x \notin \cup_{n \in \mathbb{N}} T^{-n}(\{0\})$.

The above facts show that the hyperbolic systems have an infinitesimal behavior similar to our toy model. Yet, in the arguments of the previous chapter it was essential the existence of local manifolds (lines). In the following sections we will see that hyperbolic systems do have local manifolds as well.

3.3 Invariant manifold of a fixed point

In order to gain a feeling for the type of results we will present in the next section let us consider the simplest possible case in which the existence of invariant manifolds arises: the Hadamard-Perron theorem.

Definition 3.3.1 *Given a smooth map $T : X \rightarrow X$, X being a Riemannian manifold, and a fixed point $p \in X$ (i.e. $Tp = p$) we call (local) stable manifold (of size δ) a manifold $W^s(p)$ such that⁴*

$$W^s(p) = \{x \in B_\delta(x) \subset X \mid \lim_{n \rightarrow \infty} d(T^n x, p) = 0\}.$$

Analogously, we will call (local) unstable manifold (of size δ) a manifold $W^u(p)$ such that

$$W^u(p) = \{x \in B_\delta(x) \subset X \mid \lim_{n \rightarrow \infty} d(T^{-n} x, p) = 0\}.$$

⁴Sometime we will write $W_\delta^s(p)$ when the size really matters. By $B_\delta(x)$ we will always mean the open ball of radius δ centered at x .

It is quite clear that $TW^s(p) \subset W^s(p)$ and $TW^u(p) \supset W^u(p)$ (Problem 3.8). Less clear is that these sets deserve the name “manifold.” Yet, if one thinks of the Arnold cat at the point zero (which is a fixed point) it is obvious that the stable and unstable manifolds at zero are just segments in the stable and unstable direction, the next Theorem shows that this is a quite general situation.

Theorem 3.3.2 (Hadamard-Perron) *Consider an invertible map $T : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T \in \mathcal{C}^{(1)}(U, \mathbb{R}^2)$, such that $T0 = 0$ and*

$$D_0T = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix} \quad (3.3.1)$$

where $0 < \mu < 1 < \lambda$.⁵ That is, the map T is hyperbolic at the fixed point 0. Then there exists stable and unstable manifolds at 0. Moreover, $T_0W^{s(u)}(0) = E^{s(u)}(0)$ where $E^{s(u)}(0)$ are the expanding and contracting subspaces of D_0T .

PROOF. We will deal explicitly only with the unstable manifold since the stable one can be treated exactly in the same way by considering T^{-1} instead of T .

Since the map is continuously differentiable for each $\varepsilon > 0$ we can choose $\delta > 0$ so that, in a 2δ -neighborhood of zero, we can write

$$T(x) = D_0Tx + R(x) \quad (3.3.2)$$

where $\|R(x)\| \leq \varepsilon\|x\|$, $\|D_xR\| \leq \varepsilon$.

3.3.0.b Existence—a fixed point argument

The first step is to decide how to represent manifolds. In the present case, since we deal only with curves, it seems very reasonable to consider the set of curves $\Gamma_{\delta,c}$ passing through zero and “close” to being horizontal, that is the differentiable functions $\gamma : [-\delta, \delta] \rightarrow \mathbb{R}^2$ of the form

$$\gamma(t) = \begin{pmatrix} t \\ u(t) \end{pmatrix}$$

and such that $\gamma(0) = 0$; $\|(1, 0) - \gamma'\|_\infty \leq c$. It is immediately clear that any smooth curve passing through zero and with tangent vector, at each point, in the cone $\mathcal{C} := \{(a, b) \in \mathbb{R}^2 \mid |\frac{b}{a}| \leq c\}$, can be associated to a unique element of $\Gamma_{\delta,c}$, just consider the part of the curve contained in the strip $\{(x, y) \in \mathbb{R}^2 \mid |x| \leq \delta\}$. Moreover, if $\gamma \in \Gamma_{\delta,c}$ then $\gamma \subset B_{2\delta}(0)$, provided $c \leq 1/2$.

⁵Notice that if D_0T has eigenvalues $0 < \mu < 1 < \lambda$ then one can always perform a change of variables such that (3.3.1) holds.

Notice that it suffice to specify the function u in order to identify uniquely an element in $\Gamma_{\delta,c}$. It is then natural to study the evolution of a curve through the change in the associated function.

To this end let us investigate how the image of a curve in $\Gamma_{\delta,c}$ under T looks like.

$$T\gamma(t) = \begin{pmatrix} \lambda t + R_1(t, u(t)) \\ \mu u(t) + R_2(t, u(t)) \end{pmatrix} := \begin{pmatrix} \alpha_u(t) \\ \beta_u(t) \end{pmatrix}.$$

At this point the problem is clearly that the image it is not expressed in the way we have chosen to represent curves, yet this is easily fixed. First of all, $\alpha_u(0) = \beta_u(0) = 0$. Second, by choosing $\varepsilon < \lambda$, we have $\alpha'_u(t) > 0$, that is, α_u is invertible. In addition, $\alpha_u([-\delta, \delta]) \supset [-\lambda\delta + \varepsilon\delta, \lambda\delta - \varepsilon\delta] \supset [-\delta, \delta]$, provided $\varepsilon \leq \lambda^{-1}$. Hence, α_u^{-1} is a well defined function from $[-\delta, \delta]$ to itself. Finally,

$$\left| \frac{d}{dt} \beta_u \circ \alpha_u^{-1}(t) \right| = \left| \frac{\beta'_u(\alpha_u^{-1}(t))}{\alpha'_u(\alpha_u^{-1}(t))} \right| \leq \frac{\mu c + \varepsilon}{\lambda - \varepsilon} \leq c$$

where, again, we have chosen $\varepsilon \leq \frac{c(\lambda - \mu)}{1 + c}$.

We can then consider the map $\tilde{T} : \Gamma_{\delta,c} \rightarrow \Gamma_{\delta,c}$ defined by

$$\tilde{T}\gamma(t) := \begin{pmatrix} t \\ \beta_u \circ \alpha_u^{-1}(t) \end{pmatrix} \quad (3.3.3)$$

which associates to a curve in $\Gamma_{\delta,c}$ its image under T written in the chosen representation. It is now natural to consider the set of functions $B_{\delta,c} = \{u \in \mathcal{C}^{(1)}([-\delta, \delta]) \mid u(0) = 0, |u'|_{\infty} \leq c\}$ in the vector space $Lip([-\delta, \delta])$.⁶ As we already noticed $B_{\delta,c}$ is in one-one correspondence with $\Gamma_{\delta,c}$, we can thus consider the operator $\hat{T} : Lip([-\delta, \delta]) \rightarrow Lip([-\delta, \delta])$ defined by

$$\hat{T}u = \beta_u \circ \alpha_u^{-1} \quad (3.3.4)$$

From the above analysis follows that $\hat{T}(B_{\delta,c}) \subset B_{\delta,c}$ and that $\hat{T}u$ determines uniquely the image curve.

The problem is then reduced to studying the map $\hat{T}$. The easiest, although probably not the most productive, point of view is to show that $\hat{T}$ is a contraction in the sup norm. Note that this creates a little problem since $\mathcal{C}^{(1)}$ it is not closed in the sup norm (and not even $Lip([-\delta, \delta])$ is closed). Yet, the set $B_{\delta,c}^* = \{u \in Lip([-\delta, \delta]) \mid u(0) = 0, \sup_{t,s \in [-\delta, \delta]} \frac{|u(s) - u(t)|}{|t - s|} < c\}$ is closed (see Problem 3.9). Thus $\overline{B_{\delta,c}} \subset B_{\delta,c}^*$. This means that, if we can prove that the sup norm is contracting, then the fixed point will belong to $B_{\delta,c}^*$ and we will obtain only a Lipschitz curve. We will need a separate argument to prove that the curve is indeed smooth.

⁶This are the Lipschitz functions on $[-\delta, \delta]$, that is the functions such that $\sup_{t,s \in [-\delta, \delta]} \frac{|u(s) - u(t)|}{|t - s|} < \infty$.

Let us start to verify the contraction property. Notice that

$$\alpha_u^{-1}(t) = \lambda^{-1}t + \lambda^{-1}R_1(\alpha_u^{-1}(t), u(\alpha_u^{-1}(t))),$$

thus, given $u_1, u_2 \in B_{\delta, c}$, by Lagrange Theorem

$$\begin{aligned} |\alpha_{u_1}^{-1}(t) - \alpha_{u_2}^{-1}(t)| &\leq \lambda^{-1} |\langle \nabla_\zeta R_1, (\alpha_{u_1}^{-1}(t) - \alpha_{u_2}^{-1}(t), u_1(\alpha_{u_1}^{-1}(t)) - u_2(\alpha_{u_2}^{-1}(t))) \rangle| \\ &\leq \frac{\varepsilon}{\lambda} \{ |\alpha_{u_1}^{-1}(t) - \alpha_{u_2}^{-1}(t)| + |u_1(\alpha_{u_1}^{-1}(t)) - u_2(\alpha_{u_2}^{-1}(t))| \}. \end{aligned}$$

This implies immediately

$$|\alpha_{u_1}^{-1}(t) - \alpha_{u_2}^{-1}(t)| \leq \frac{\lambda^{-1}\varepsilon}{1 - \lambda^{-1}\varepsilon} \|u_1 - u_2\|_\infty. \quad (3.3.5)$$

On the other hand

$$\begin{aligned} |\beta_{u_1}(t) - \beta_{u_2}(t)| &\leq \mu |u_1(t) - u_2(t)| + |\langle \nabla_\zeta R_2, (0, u_1(t) - u_2(t)) \rangle| \\ &\leq (\mu + \varepsilon) \|u_1 - u_2\|_\infty. \end{aligned} \quad (3.3.6)$$

Moreover,

$$|\beta'_u(t)| \leq \mu + \varepsilon. \quad (3.3.7)$$

Collecting the estimates (3.3.5, 3.3.6, 3.3.7) readily yields

$$\begin{aligned} \|\hat{T}u_1 - \hat{T}u_2\|_\infty &\leq \|\beta_{u_1} \circ \alpha_{u_1}^{-1} - \beta_{u_1} \circ \alpha_{u_2}^{-1}\|_\infty + \|\beta_{u_1} \circ \alpha_{u_2}^{-1} - \beta_{u_2} \circ \alpha_{u_2}^{-1}\|_\infty \\ &\leq \left\{ [\mu + \varepsilon] \frac{\lambda^{-1}\varepsilon}{1 - \lambda^{-1}\varepsilon} + (\mu + \varepsilon) \right\} \|u_1 - u_2\|_\infty \\ &\leq \sigma \|u_1 - u_2\|_\infty, \end{aligned}$$

for some $\sigma \in (0, 1)$, provided ε is chosen small enough.

Clearly, the above inequality immediately implies that there exists a unique element $\gamma_* \in \Gamma_{\gamma, c}$ such that $\hat{T}\gamma_* = \gamma_*$, this is the *local* unstable manifold of 0.

3.3.0.c Regularity—a cone field

As already mentioned, a separate argument it is needed to prove that γ_* is indeed a $\mathcal{C}^{(1)}$ curve.

To prove this, one possibility would be to redo the previous fixed point argument trying to prove contraction in $\mathcal{C}_{Lip}^{(1)}$ (the $\mathcal{C}^{(1)}$ functions with Lipschitz derivative); yet this would require to increase the regularity requirements on T . A more geometrical, more instructive and more inspiring approach is the following.

Define the cone field $\mathcal{C}_{\theta,h}(x, u) := \{\xi \in B_h(x) \mid (a, b) = \xi - x; a \neq 0; |\frac{b}{a} - u| \leq \theta\}$, with $|u| \leq c\delta$, $\theta \leq c\delta$ and $h \leq \delta$. By construction $B_h(x) \cap \gamma_* \subset \mathcal{C}_{c\delta,h}$ for each $x \in \gamma_*$. We will study the evolution of such a cone field on γ_* .

For all $\xi \in \mathcal{C}_{\theta,h}(x, u)$, if $(a, b) = \xi - x$ and $(\alpha, \beta) = T\xi - Tx$, it holds

$$(\alpha, \beta) = D_x T(a, b) + \mathcal{O}(C\|(a, b)\|^2).$$

Thus, setting $(\alpha', \beta') = D_x T(a, b)$ and $u' = \frac{\beta'}{\alpha'}$,

$$\left| \frac{\beta}{\alpha} - u' \right| \leq \mu\lambda^{-1}[c_1 h + \theta],$$

for some constant c_1 depending only on T and δ . Accordingly, if h is small enough, there exists $\sigma \in (0, 1)$ such that

$$T\mathcal{C}_{\theta,h}(x, u) \subset \mathcal{C}_{\sigma\theta,h}(Tx, u').$$

Hence, if $x \in \gamma_*$, $\gamma_* \cap B_{\sigma^{-n}h}(T^{-n}x) \subset \mathcal{C}_{c\delta,h}(T^{-n}x, 0)$ and, since $T^{-n}\gamma_* \subset \gamma_*$,

$$\gamma_* \cap B_h(x) \subset \mathcal{C}_{\sigma^n c, h}(x, v_n) \quad (3.3.8)$$

where $(a, av_n) = D_{T^{-n}x} T^n(1, 0)$ and $h < \sigma^n c\delta$.

The estimate (3.3.8) clearly implies

$$\gamma'_*(x) = (1, \lim_{n \rightarrow \infty} v_n) \quad (3.3.9)$$

which indeed exists (see Problem 3.10). $\square$

There is an issue not completely addresses in our formulation of Hadamard-Perron theorem: the uniqueness of the manifolds.⁷ It is not hard to prove that $W^{s(u)}(p)$ are indeed unique (see Problem 3.11).

The point of view employed in the previous theorem is brought to its extreme consequences in [46].

There is another point of view that can be adopted in the study of stable and unstable manifolds: to “grow” the manifolds. This is done by starting with a very short curve in $\Gamma_{\delta,c}$, e.g. $\gamma_0(t) = (t, 0)$ for $t \in [\lambda^{-n}\delta, \lambda^n\delta]$, and showing that the sequence $\gamma_n := T^n\gamma_0$ converges to a curve in the strip $[-\delta, \delta]$, independent of γ_0 . From a mathematical point of view, in the present case, it corresponds to spell out explicitly the proof of the fixed point theorem. Nevertheless, as we will see in ???, it is a more suggestive point of view and it is more convenient when the hyperbolicity is non uniform. For example consider the map⁸

$$T \begin{pmatrix} x \\ y \end{pmatrix} := \begin{pmatrix} 2x - \sin x + y \\ x - \sin x + y \end{pmatrix} \quad (3.3.10)$$

⁷Namely the doubt may remain that a less regular set satisfying Definition 3.3.1 exists.

⁸Some times this is called *Lewowicz map*, see [1].

then 0 is a fixed point of the map but

$$D_0T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not hyperbolic, yet, due to the higher order terms, there exist stable and unstable manifolds (see Problems 3.12, 3.13, 3.14).

3.4 Invariant manifolds and foliations (existence)

The concept of stable and unstable manifold of a fixed point can be obviously generalized to periodic orbits and, less obviously but more interestingly, to any orbit:

Definition 3.4.1 *For each point $x \in X$ we define its stable (unstable) sets as*

$$W^{s(u)}(x) = \{z \in X \mid \lim_{n \rightarrow \infty(-\infty)} d(T^n x, T^n z) = 0\}.$$

Remark 3.4.2 *It will not surprise the reader that very often such sets are indeed manifolds, in this case they are called stable (unstable) manifolds. In the following we will discuss only this case.*

Very often the structure of the stable manifolds is quite complex (as we have already seen in the introduction) and for this reason it may be more convenient to talk about *local* stable (unstable) manifolds as we did already (see Definition 3.3.1).

Definition 3.4.3 *By local stable (unstable) manifolds at $x \in X$ we mean the connected component containing x of $B_\varepsilon(x) \cup W^{s(u)}(x)$, for some $\varepsilon > 0$.*

Clearly the local manifolds are never unique, since their size may vary, but they may be unique once the size is assigned. Sometimes local stable manifolds will be denoted by $W_{\text{loc}}^s(x)$ as well as $W_\delta^s(x)$, we will use the former only if some confusion can arise and the latter only if the size plays an explicit rôle in the argument.

Definition 3.4.4 *By $W_\delta^s(x)$ we mean a local stable manifold such that $\partial W_\delta^s(x) \subset \partial B_\delta(x)$.*

For simplicity we will restrict ourself to stable manifolds until the end of the section, note that if the map is invertible, then the same considerations hold for the unstable manifold as well.

We will be mostly interested in the case in which almost every point has a stable manifold. Note that if $W^s(x)$ is a local stable manifold at x , then $TW^s(x)$ is a local stable manifold at Tx . This means that we have an invariant foliation

Definition 3.4.5

✕ define foliation, cite KH

The first interesting fact is the following.

Proposition 3.4.6 *If we have a stable foliation, then for almost all points $x \in X$ the associated stable manifold contracts exponentially. That is there exists $K \in \mathbb{R}^+$, $\nu \in (0, 1)$:*

$$d(T^n \xi, T^n \eta) \leq K \nu^n \ell(x) \quad \xi, \eta \in W_{\text{loc}}^s(x),$$

where $\ell(x)$ is the size of the local stable manifold at x .

PROOF. Let $\Lambda_\delta = \{x \in X \mid x \text{ has stable manifold of size } \delta\}$, by hypothesis

$$m(\cup_{k \in \mathbb{N}} \Lambda_{\frac{1}{k}}) = m(X).$$

Moreover, by definition, for each $\delta > 0$ and $x \in \Lambda_\delta$ there exists $n_\delta(x) \in \mathbb{N}$ such that

$$d(T^n \xi, T^n \eta) < \frac{1}{2} \ell(x) \quad \forall \xi, \eta \in W_\delta^s(x), \quad n > n_\delta(x).$$

Let us then define $\Lambda_{\delta,L} := \{x \in \Lambda_\delta \mid n_\delta(x) \leq L\}$. Clearly $m(\cup_{L \in \mathbb{N}} \Lambda_{\delta,L}) = m(\Lambda_\delta)$ for each $\delta > 0$. By the above remarks we can choose δ, L such that $m(\Lambda_{\delta,L}) > 0$. We start by considering the *separated* return map $T_{\delta,L}(x) := T^{n(\delta,L,x)} x$ to the set $\Lambda_{\delta,L}$ (see 1.2) where $n(\delta, L, x) := \min\{n \geq L : T^n x \in \Lambda_{\delta,L}\}$. It is then natural to define the return times $\{n_j(x)\}$, $x \in \Lambda_{\delta,L}$, such that $T^{n_j(x)} x = T_{\delta,L}^j x$. Thus, for each $n \in \{n_j(x), \dots, n_{j+1}(x) - 1\}$ holds

$$d(T^n \xi, T^n \eta) \leq 2^{-j} \ell(x). \quad (3.4.1)$$

Lemma 3.4.7 *Given $U \subset X$ such that (3.4.1) holds, almost every $x \in U$ has a manifold $W_\delta(x)$ that contracts exponentially.*

PROOF. Let $j_n(x) := \frac{1}{n} \sum_{i=1}^n \chi_U(T^i x)$, then (3.4.1) reads

$$d(T^n \xi, T^n \eta) \leq 2^{-n j_n(x)} \ell(x)$$

but Birkhoff Theorem implies that $j_n \rightarrow j^+$ almost everywhere and that $A_0 := \{x \in U \mid j^+(x) = 0\}$ has measure zero (see Problem .) thus for almost all $x \in U$ there exists $\bar{n}$ such that $j_n(x) \geq \frac{1}{2} j^+(x)$ for all $n \geq \bar{n}$. From this it follows immediately that there exists $K > 0$:

$$d(T^n \xi, T^n \eta) \leq K 2^{-\frac{n}{2} j^+(x)} \ell(x)$$

□

□

The use of non-invariant sets on which the systems has some useful property—on which the previous proof is based—reaches its glorification in the notion of *Pesin sets* which are the key ingredioend in the proof of Theorem 3.4.8. In fact, the importance of the notion of hyperbolicity lies almost entirely in the following result.⁹

Theorem 3.4.8 (Pesin [71]) *Let (X, μ, T) be a Smooth Hyperbolic System, almost each point $x \in X$ has a local stable manifolds $W_\varepsilon^s(x)$, for some $\varepsilon > 0$, that is a manifold with the following properties:*

1. $x \in W_\varepsilon^s(x)$;
2. $T_x W_\varepsilon^s(y) = E^s(y) \quad \forall y \in W_\varepsilon^s(x)$;
3. *there exists $K > 0$, and $\nu \in (0, 1)$ such that, for each $y \in W_\varepsilon^s(x)$*
 $d(T^n x, T^n y) \leq K \nu^n d(x, y)$.

and such manifolds are unique.

Note that, if T is invertible, the previous theorem applied to T^{-1} yield a local unstable manifold. Moreover, if $W^s(x)$ is a local stable manifold at x , then there must exist local stable manifolds at Tx as well (since $TW^s(x)$ is one of them). From now on we will assume that W^s are the local stable manifolds provided by Pesin Theorem, without entering in a real discussion concerning their size, which can be proven to enjoy some extremely mild form of uniformity.

We will not prove 3.4.8 in its full generality, but see 7 for a proof in a less general setting.

Definition 3.4.9 *At each point x for which the local stable manifold exists we can define the global stable manifold has*

$$\overline{W}^s(x) = \bigcup_{n=0}^{\infty} T^{-n} W^s(T^n x).$$

Remark 3.4.10 *One can then see that the general situation is not too dissimilar from the two dimensional automorphism of the torus, in particular, if the system is Anosov, then $\overline{W}^s$ turns out to be of infinite size. Yet, if the*

⁹To appreciate better the next theorem consider that the Oseledets Theorem provides invariant distributions made of stable and unstable subspaces, then Pesin theorem essentially says that such distributions are integrable. In general, to integrate a distribution are necessary both regularities and geometrical properties (crf. Frobenius Theorem [14]) that, typically, are not verified (regularity) and very hard to verify (geometry) in the present setting. Only the fact that the distribution is of a dynamical origin saves the day.

system is non-uniformly hyperbolic, then $\overline{W^s}$ can be arbitrarily short or it may be long in its own metric but wiggle so much as to be contained in a very small ball.

3.5 Invariant manifolds and foliations (regularity)

The only other property that was really used in chapter 2 was the possibility to use the stable and unstable foliation as the basis for new coordinates. The relevant property, since we are dealing with measures, is that the change of coordinates does not substantially alter the measure: that is the new measure (obtained by the invariant one via the change of coordinates) is absolutely continuous with respect to the invariant one. If we want to extend the applicability of such an argument we need some analogous property. The right generalization turns out to be the so called *absolute continuity*. To describe it let us consider $x \in X$ with $W^s(x) \neq \emptyset$. Then, for a.e. such x , there exists a sufficiently small neighborhood $U(x)$ such that, for any two manifolds W_1, W_2 transversal to $W^s(x)$, the set $A = \{z \in W_1 \cap U(x) \mid W^s(z) \cap W_2 \neq \emptyset\}$ is not empty. In addition, if by μ_{W_i} we mean the measure induced on W_i by the Riemannian metric, $\mu_{W_1}(A) > 0$. Let us define $\phi : A \rightarrow W_2$ by $\phi(z) = W^s(z) \cap W_2$. We state the following in the simplest case.

Theorem 3.5.1 (Pesin [71]) *If the system (X, μ, T) is smooth, hyperbolic and μ is the Riemannian volume, then, for almost all $x \in X$ the measure $\phi_*\mu_{W_1}$ is absolutely continuous with respect to the measure μ_{W_2} .*

In our example we had the extreme case $\phi_*\mu_{W_1} = \mu_{W_2}$, yet it turns out that the above weaker property suffices to push the Hopf argument through, as we will see in chapters 8, 9. For the time being let us only quote the following general fact:

Theorem 3.5.2 ([71],[60]) *If the system (X, μ, T) is smooth, hyperbolic and μ is the Riemannian volume, then it has, at most, countably many ergodic components.¹⁰*

✕ Put general results on foliation regularity from [BKL]

¹⁰An ergodic component is a set which is the support of an invariant ergodic measure. The ergodic components of a measure are the measures associated to its ergodic decomposition. In other words μ can be written as the convex combination of at most countably many ergodic measures.

3.6 Comments on the non-smooth case

The results of the last section, although quite general, have a shortcoming: the smoothness requirements. As we will see in the following, systems that are quite natural both from the mathematical point of view and from the physical one are not smooth—typically they have discontinuities. In this section we will discuss a class of systems called *smooth systems with singularities*. Although the theory of such systems has been done in great generality, here we will give a restrictive definition, just sufficient for our later purposes. See the notes at the end of the chapter for information on more general settings.

Definition 3.6.1 *By Smooth Dynamical System with singularities we mean a Dynamical Systems (X, T, μ) , where*

- *X is the union of finitely many compact pieces X_i of $\mathbb{R}^n$, ∂X_i is the union of finitely many $n - 1$ dimensional smooth manifolds.*
- *T is smooth outside a compact set $\mathcal{S}$. The singularity set $\mathcal{S}$ is the finite union of smooth $n - 1$ dimensional manifolds with boundary $\mathcal{S}_i$, $\mathcal{S}_i \cup \mathcal{S}_j \not\subset \partial \mathcal{S}_i \cap \partial \mathcal{S}_j$ implies $i = j$. In addition, the boundary $\partial \mathcal{S}_i$ is the finite union of smooth $n - 2$ dimensional manifolds.*
- *There exists $c_1, c_2 > 0$ such that*

$$\|D_x T\| + \|D_x^2 T\| + \|D_x T^{-1}\| + \|D_x^2 T^{-1}\| \leq c_1 \text{dist}(x, \mathcal{S})^{-c_2}.$$

- *The measure μ is absolutely continuous with respect to Lebesgue.*

Remark 3.6.2 *Note that the fact that (X, T, μ) is a Smooth Dynamical System with singularities does not implies immediately that the same holds for (X, T^k, μ) . The problem is that the map T can be very wild near the set $\mathcal{S}$, so it is not clear that the singularity set of T^k will satisfy our requirements. Nevertheless, in the examples we will consider, all the Dynamical System (X, T^k, μ) will always be Smooth Dynamical System with singularities.*

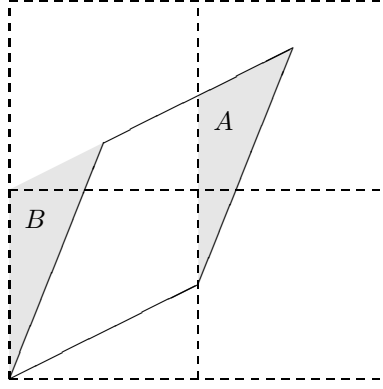
Remark 3.6.3 *We will call a smooth Dynamical System with singularities invertible if T^{-1} is densely defined and (X, T^{-1}, μ) is itself a smooth Dynamical System with singularities.*

Note that the above conditions imply the applicability of Oseledec Theorem (Problem ???).

3.6.1 Examples

3.6.1.a Backer map

It is easy to check that the Backer map is a Smooth Dynamical System with singularities.

Figure 3.1: Definition of the map T

3.6.1.b Discontinuous Arnold cat

If we consider $(\mathbb{R}^2, T_0, m)$ defined by

$$T_0 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} =: A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad (3.6.1)$$

with $a \notin \mathbb{Z}$, then it is not possible to quotient the system down to a torus preserving the continuity of the map. Yet, it is possible to construct a discontinuous version of the Arnold cat. To be more concrete, let us consider the case $a = 1/2$,

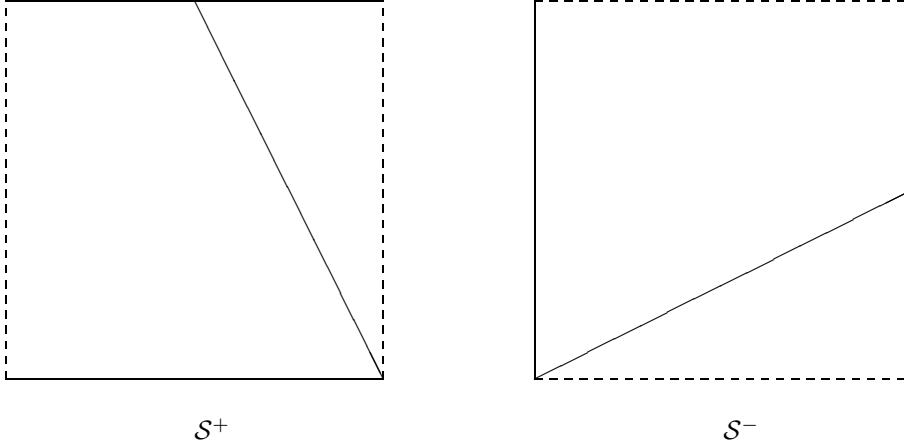
$$A = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{5}{4} \end{pmatrix}. \quad (3.6.2)$$

To define a map on the torus we must specify how to restrict T_0 to the torus. To do so we consider the fundamental domain $D = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1; 0 \leq y \leq 1\}$. Figure 3.1 shows the image $T_0 D \subset \mathbb{R}^2$.

To define a map on the two dimensional torus $\mathbb{T}^2$ it suffices to cut the shaded region A from the image $T_0 D$ and translate it over the shaded region B . After that one identifies the points $\mod 1$ as usual. We will call $T : \mathbb{T}^2 \rightarrow \mathbb{T}^2$ the map just defined. The discontinuities of T are the two lines $\{y = 0\}$ and $\{y = 2(1 - x)\}$. The explicit formula for the map $(x', y') = T(x, y)$ reads

$$\begin{aligned} x' &= x + \frac{y \mod 1}{2} \mod 1 \\ y' &= \frac{1}{2}(x + \frac{y}{2} \mod 1) + y \mod 1. \end{aligned} \quad (3.6.3)$$

We leave to the reader to check that the above formulae yields precisely the map described in Figure 3.1.

Figure 3.2: Discontinuity lines of the maps T and T^{-1}

The stable and unstable directions can still be found by diagonalizing the matrix A . The two eigenvalues are $\lambda = (9 + \sqrt{17})/8$ and $\lambda^{-1} = (9 - \sqrt{17})/8$ while the unstable direction is given by $v^u = (1, (1 + \sqrt{17})/2)$ and the stable by $v^s = (1, (1 - \sqrt{17})/2)$. This means that the angle between the unstable direction and the vertical part of $\mathcal{S}^-$ is about 38 degrees, while the angle between v^u and the diagonal part of $\mathcal{S}^-$ is about 25 degrees (see figure 3.2).

The status of the stable and unstable manifolds is more complex. A line parallel to, say, the unstable direction is cut by $\mathcal{S}^-$ into pieces and if y and z belong to two different pieces the distance $d(T^n y, T^n z)$ does not decrease to zero as $n \rightarrow -\infty$. Since this last property is the definition of unstable manifold (and is of crucial importance in the Hopf method), the unstable (and stable) leaves are going to be much shorter than in the smooth case. Here is how we construct them. We will formulate everything for the unstable leaves alone, the construction of the stable leaves being completely similar.

We proceed inductively. Thus, for $x \in \text{int}(\mathcal{M}^-)$, we define $W_1^u(x)$ as the open segment of the line through x with the direction of the unstable eigenvector which contains x and has both endpoints on $\mathcal{S}^-$. The preimage $T^{-1}W_1^u(x)$ is by a factor of λ shorter than $W_1^u(x)$ and, in general, is cut into two, three or four pieces by $\mathcal{S}^-$.¹¹ We pick the piece which contains $T^{-1}x$ and take its image under T ; this is our second approximate unstable leaf $W_2^u(x)$, i.e.,

$$W_2^u(x) = T \left(T^{-1}W_1^u(x) \cap W_1^u(T^{-1}(x)) \right).$$

¹¹It can be cut in three pieces only if it is sufficiently close to $(0,0)$ or $(1,1/2)$, while it can be cut in four only if it is long enough to be close to both points simultaneously. This last possibility can take place only for $T^{-1}W_1^u(x)$, the connected pieces of $T^{-2}W_1^u(x)$ being already too short (see figure 3.2).

Unless $T^{-1}x \in \mathcal{S}^-$ the second approximate unstable leaf $W_2^u(x)$ is again an open segment containing x with endpoints on $\mathcal{S}^- \cup T\mathcal{S}^-$ and naturally $W_2^u(x) \subset W_1^u(x)$. By construction T^{-2} has no discontinuities on $W_2^u(x)$. Given $W_n^u(x)$, $n = 1, 2, \dots$, we define the $n + 1$ approximate unstable leaf of x , $W_{n+1}^u(x)$, by

$$W_{n+1}^u(x) = T^n (T^{-n}W_n^u(x) \cap W_1^u(T^{-n}(x))) .$$

If $x \notin \bigcup_{i=0}^{+\infty} T^i \mathcal{S}^-$ then this inductive procedure will yield a nested sequence of open segments containing x

$$W_1^u(x) \supset W_2^u(x) \supset \dots$$

with endpoints on

$$\bigcup_{i=0}^{+\infty} T^i \mathcal{S}^- .$$

We can also describe this construction in the following way. First we consider a fairly long segment $W_1^u(x)$. Then we look at $T\mathcal{S}^-$, if it does not intersect $W_1^u(x)$ then we do not change it, if it splits $W_1^u(x)$ into several segments, then we keep the segment which contains x . We repeat it with $T^2\mathcal{S}^-$ and further images of $\mathcal{S}^-$, so that the segment may be cut shorter infinitely many times. The property $x \notin \bigcup_{i=0}^{+\infty} T^i \mathcal{S}^-$ ensures that x stays always strictly inside the segment. It is quite remarkable that, for almost every x , this inductive process shortens the segment only finitely many times. More precisely we have

Proposition 3.6.4 *For almost all $x \in \mathbb{T}^2 \setminus \bigcup_{i=0}^{+\infty} T^i \mathcal{S}^-$ the sequence of approximate unstable leaves of x stabilizes, i.e., there is a natural $N = N(x)$ such that*

$$\bigcap_{i=1}^{+\infty} W_i^u(x) = \bigcap_{i=1}^N W_i^u(x).$$

PROOF. For $t > 0$, let

$$X_t = \{x \in \mathbb{T}^2 \mid d(x, \mathcal{S}^-) \leq t\}$$

where $d(\cdot, \cdot)$ is the distance of a point from a set. Because $\mathcal{S}^-$ is the finite union of three segments we have

$$\mu(X_t) \leq 3t.$$

Choosing $t_n = \frac{1}{n^2}$ we get

$$\sum_{n=1}^{+\infty} \mu(X_{t_n}) < +\infty,$$

hence also

$$\sum_{n=1}^{+\infty} \mu(T^n X_{t_n}) < +\infty.$$

It follows by the Borel-Cantelli Lemma that almost every x belongs to only finitely many of the sets ✖ Make references or do problem

$$TX_{t_1}, T^2X_{t_2}, \dots,$$

which means that except for finitely many values of n

$$d(T^{-n}x, \mathcal{S}^-) > \frac{1}{n^2}.$$

Choosing $c(x) > 0$ sufficiently small we can take care of the finite number of exceptional values of n so that

$$d(T^{-n}x, \mathcal{S}^-) > \frac{c(x)}{n^2}$$

for each $n = 1, 2, \dots$. Each time $W_{n+1}^u(x)$ is shorter than $W_n^u(x)$ we must have

$$d(T^{-n}x, \mathcal{S}^-) < \frac{\text{length}(W_n^u(x))}{\lambda^n}.$$

But then

$$\frac{c(x)}{n^2} < \frac{\text{length}(W_n^u(x))}{\lambda^n} \leq \frac{\text{length}(W_1^u(x))}{\lambda^n} \leq 2\lambda^{-n},$$

which can hold for at most finitely many values of n . □

We define the unstable leaf only for points x in the set of full measure described in Proposition 4.1, by taking the intersection

$$W^u(x) = \bigcap_{i=1}^{+\infty} W_i^u(x).$$

In view of Proposition 3.6.4, for each $W^u(x)$, there are natural numbers $n_l(x)$ and $n_r(x)$ such that $T^{n_l(x)}W^u(x)$ has the left endpoint on $\mathcal{S}^-$ and $T^{n_r(x)}W^u(x)$ has the right endpoint on $\mathcal{S}^-$. Most importantly we have the exponential contraction of $W^u(x)$, i.e., for $y \in W^u(x)$ the distance

$$d(T^{-n}y, T^{-n}x) = \frac{d(y, x)}{\lambda^n} \rightarrow 0 \text{ as } n \rightarrow +\infty.$$

Everything that we have done to construct the unstable leaves can be repeated for the stable leaves and they have analogous properties.

This it is encouraging, yet it is clearly not sufficient to perform the Hopf argument. We will discuss this further and with much more care in chapter 9. For the time being it suffices to notice that what we have seen so far implies that the discontinuous Arnold cat has, at most, countably many ergodic components (see Problem counter).

The interest of the Smooth Dynamical System with singularities is that most of the theory of the previous sections holds for this more general systems. Namely Theorems 3.4.8, 3.5.1, [52].

3.7 Flows

All what we have described so far has a rather straightforward generalization in the case of flows, yet some natural changes are called for.

To appreciate the problem let us consider a flow, on a compact Riemannian manifold, generated by a smooth non-zero vector field V . By definition $\frac{d}{dt}\phi^t|_{t=0} = V(x)$ and $d\phi^t V(x) = V(\phi^t x)$, thus $\lambda(x, V(x)) = 0$. This is a rather general fact: the Lyapunov exponents in the flow direction is zero. The only relevant exception is constituted by hyperbolic fixed points (think of the unstable equilibrium point of the pendulum) that, in the previous example, was ruled out by the assumption that the vector field be non zero. We will consider only such case.

Consequently a flow is hyperbolic if the tangent space is split in three transversal subspaces E^s, E^u, E^0 , where E^0 is the flow direction and corresponds to a zero Lyapunov exponents.

Oseledets Theorem (Theorem 3.2.2) holds unchanged with (3.2.1) obviously replaced by

$$\int_X \|\log d\phi^t\| d\mu < \infty.$$

For a smooth flow coming from a non vanishing vector field Theorem 3.2.3 holds unchanged as well, the same for Theorems 3.4.8, 3.5.1 and 3.5.2.

The extension to non-smooth cases require a bit of care. we will do it only for the symplectic case (Hamiltonian flows with collisions), which is the focus of the following chapters.

3.7.1 Examples

3.7.1.a Smooth flows with collisions

Let M be a smooth manifold with piecewise smooth boundary ∂M . We assume that the manifold M is equipped with a symplectic structure ω .¹² Given a smooth function H on M with *non vanishing* differential we obtain the non vanishing

¹²That is a non-degenerate closed antisymmetric two form.

Hamiltonian vector field $F = \nabla_\omega H$ on M by $\omega(\nabla_\omega H, v) = dH(v)$. The vector field F is tangent to the level sets of the Hamiltonian $M^c = \{z \in M | H(z) = c\}$.

We distinguish in the boundary ∂M the regular part, ∂M_r , consisting of the points which do not belong to more than one smooth piece of the boundary and where the vector field F is transversal to the boundary. The regular part of the boundary is further split into “outgoing” part, ∂M_- , where the vector field F points outside the manifold M and the “incoming” part, ∂M_+ , where the vector field is directed inside the manifold. Suppose that additionally we have a piecewise smooth mapping $\Gamma : \partial M_- \rightarrow \partial M_+$, called the collision map. We assume that the mapping Γ preserves the Hamiltonian, $H \circ \Gamma = H$, and so it can be restricted to each level set of the Hamiltonian.

We assume that all the integral curves of the vector field F that end (or begin) in the singular part of the boundary lie in a codimension 1 submanifold of M .

We can now define a flow $\Psi^t : M \rightarrow M$, called a flow with collisions, which is a concatenation of the continuous time dynamics Φ^t given by the vector field F , and the collision map Γ . More precisely a trajectory of the flow with collisions, $\Psi^t(x)$, $x \in M$, coincides with the trajectory of the flow Φ^t until it gets to the boundary of M at time $t_c(x)$, the collision time. If the point on the boundary lies in the singular part then the flow is not defined for times $t > t_c(x)$ (the trajectory “dies” there). Otherwise the trajectory is continued at the point $\Gamma(\Psi^{t_c}x)$ until the next collision time, i.e., for $0 \leq t \leq t_c(\Gamma(\Psi^{t_c}x))$

$$\Psi^{t_c+t}x = \Phi^t \Gamma \Psi^{t_c}x.$$

We define a flow with collisions to be symplectic, if for the collision map Γ restricted to any level set M^c of the Hamiltonian we have

$$\Gamma^* \omega = \omega,$$

for some non vanishing function β defined on the boundary. More explicitly we assume that for every vectors ξ and η from the tangent space $T_z \partial M^c$ to the boundary of the level set M^c we have

$$\omega(D_z \Gamma \xi, D_z \Gamma \eta) = \omega(\xi, \eta).$$

We restrict the flow with collisions to one level set M^c of the Hamiltonian and we denote the resulting flow by Ψ_c^t . This flow is very likely to be badly discontinuous but we can expect that for a fixed time t the mapping Ψ_c^t is piecewise smooth, so that the derivative $D\Psi_c^t$ is well defined except for a finite union of codimension one submanifolds of M^c . We will consider only such cases.

The symplectic volume $\wedge^d \omega$ is clearly invariant for the flow, so will be the measure μ_c obtained by restricting the symplectic volume to the manifold M^c . Clearly for such an invariant measure all the trajectories that begin (or end) in the singular part of the boundary have measure zero. With respect to the measure μ_c

the flow Ψ_c^t is a measurable flow in the sense of Definition 1.1.3 and we obtain a measurable derivative cocycle $D\Psi_c^t : T_x M^c \rightarrow T_{\Psi_c^t x} M^c$. We can define Lyapunov exponents of the flow Ψ_c^t with respect to the measure μ_c , if we assume that

$$\begin{aligned} \int_{M^c} \log_+ \|D_x \Psi_c^t\| d\mu_c(x) &< +\infty \\ \int_{\partial M_-^c} \log_+ \|D_y \Gamma\| d\mu_{cb}(y) &< +\infty \end{aligned} \quad (3.7.1)$$

✘ Check why the second condition is needed, is not the first enough?

(cf.[69],[76]).

Problems

3.1 Show that if a system is uniformly hyperbolic, then the distributions are uniformly transversal. (Hint: Let $v \in E^u$ and $w \in E^s$, $\|v\| = \|w\| = 1$, and let $\cos \theta := \langle v, w \rangle$ be the angle between them.

$$\|D_x T^n\| \theta^2 \geq \|D_x T^n\| 2(1 - \cos \theta) \geq \|D_x T^n(v - w)\| \geq c^{-1} \nu^n - c \nu^n.$$

the result follows by choosing n large enough and by the continuity of $D_x T$.)

3.2 Prove that $\lambda(Tx, D_x T v) = \lambda(x, v)$.

3.3 Prove that $\lambda(x, v + w) \leq \max\{\lambda(x, v), \lambda(x, w)\}$ and $\lambda(x, \alpha v) = \lambda(x, v)$ for each $\alpha \in \mathbb{R}$, if they all exist. (Hint: Just apply the definition of LE and note that

$$\lambda(x, v + w) \leq \lim_{n \rightarrow \infty} \max\left\{\frac{1}{n} \log \|D_x T^n v\|, \frac{1}{n} \log \|D_x T^n w\|\right\}.$$

)

3.4 Assuming only that the LE are well defined a.e., prove that, if (X, T, μ) is ergodic, X is a d dimensional manifold and T a diffeomorphism, then there exists d numbers $\{\lambda_i\}$ such that the Lyapunov exponents $\lambda(x, v) \in \{\lambda_i\}$ a.e.. (Hint: For each $\alpha \in \mathbb{R}$ define $V_\alpha(x) := \{v \in T_x X \mid \lambda(x, v) \leq \alpha\}$. By Problem 3.3 $V_\alpha(x)$ is a linear vector space and, by Problem 2 the distribution V_α is invariant. Then $d_\alpha(x) := \dim V_\alpha(x)$ is an invariant function, thus a.e. constant for each α . In addition, d_α is an increasing function of α and can assume only the values $\{0, \dots, d\}$. Thus there are at most $s \leq d$ $\{\alpha_j\}$ where d_α jumps. But this means that the LE are discrete. In fact, let $v \in V_\alpha(x) \setminus V_\beta(x)$, $\alpha > \beta$, then for each $w \in \text{span}\{v, V_\beta(x)\}$ it is easy to compute that $\lambda(x, w) = \lambda(x, v) > \beta$,

which means: the LE is constant over $V_\alpha(x)$ apart for lower dimensional subspaces. In addition, we have a flag of subspaces $\{V_i\}_{i=0}^s$, $s \leq d$, such that $V_\alpha \in \{V_i\}_{i=0}^s$ for each $\alpha \in \mathbb{R}$. Hence, if $V_\alpha \supset V_i$ but $V_\alpha \not\supset V_{i+1}$ it must be $V_\alpha = V_i$, thus if $v \in V_\alpha$ but $v \notin V_{i-1}$ $\lambda(x, v) = \alpha_i$ where $\alpha_i = \inf\{\alpha \in \mathbb{R} \mid V_\alpha \supset V_i\}$.)

3.5 Show that, if T is invertible, $\{\lambda_i(x)\}$ is equal a.e. to $\{-\lambda_i^-(x)\}$ where $\{\lambda_i^-(x)\}$ are the LE of (X, T^{-1}, μ) .

3.6 Show that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log |\det(D_x T^n)|$$

exists almost everywhere. (Hint: Apply BET.)

3.7 Let (X, T, μ) be a Dynamical Systems, X a compact Riemannian manifold and T a.e. differentiable. Suppose that there exists a one-dimensional distribution $E(x)$ such that $D_x T E(x) = E(Tx)$. Prove, without using Oseledets theorem, that for each $v \in E(x)$ the LE $\lambda(x, v)$ is well defined. (Hint: Let $v(x) \in E(x)$, $\|v(x)\| = 1$, then $D_x T v(x) = \alpha(x)v(Tx)$ and thus $D_x T^n v(x) = \prod_{i=1}^n \alpha(T^i x)v(T^i x)$. Then the result follows by the BET.)

3.8 Show that, if p is a fixed point, then $TW^s(p) \subset W^s(p)$ and $TW^u(p) \supset W^u(p)$.

3.9 Prove that the set $B_{\delta, c}^*$ in section 3.3 is closed with respect to the sup norm $\|u\|_\infty = \sup_{t \in [-\delta, \delta]} |u(t)|$.

3.10 Prove that the limit in (3.3.9) is well defined.

3.11 Prove that, in the setting of Theorem 3.3.2, the unstable manifold is unique. (Hint: This amounts to show that the set of points that are attracted to zero are exactly the manifolds constructed in Theorem 3.3.2. Use the local hyperbolicity to show that.)

3.12 Consider the Levowich map (3.3.10), show that, given the set of curves $\Gamma_{\delta, c} := \{\gamma : [-\delta, \delta] \rightarrow \mathbb{R}^2 \mid \gamma(t) = (t, u(t)); \gamma(0) = 0; |u'(t)| \in [c^{-1}t, ct]\}$, it is possible to construct the map $\tilde{T} : \Gamma_{\delta, c} \rightarrow \Gamma_{\delta(1+c^{-1}\delta), c}$ in analogy with (3.3.3).

3.13 In the case of the previous problem show that for each $\gamma_i \in \Gamma_{\delta, c}$ holds $d(\tilde{T}\gamma_1, \tilde{T}\gamma_2) \leq (1 - c\delta)d(\gamma_1, \gamma_2)$.

✘ NO! change according to [LM]

3.14 Show that for the Levowich map zero has a unique unstable manifold. (Hint: grow the manifolds, that is, for each $n > 1$ define $\delta_n := \frac{\rho}{n}$. Show

that one can choose ρ such that $\delta_{n-1} \geq \delta_n(1 + c^{-1}\delta_n)$. according to Problem 12 it follows that $\tilde{T} : \Gamma_{\delta_n, c} \rightarrow \Gamma_{\delta_{n-1}, c}$. Moreover,

$$d(\tilde{T}^{n-1}\gamma_1, \tilde{T}^{n-1}\gamma_2) \leq \prod_{i=1}^n n(1 + c^{-1}\delta_i)d(\gamma_1, \gamma_2).$$

Finally, show that, setting $\gamma_n(t) = (0, t) \in \Gamma_{\delta_n, c}$, the sequence $\tilde{T}^{n-1}\gamma_n$ is a Cauchy sequence that converges in $\mathcal{C}^{(0)}$ to a curve in $\Gamma_{1, c}$ invariant under $\tilde{T}$.)

- 3.15** Prove that Oseledets Theorem can be applied to smooth Dynamical Systems with singularities. (Hint: Just check the (3.2.1) is satisfied.)
- 3.16** Prove that the discontinuous Arnold cat (Examples ??) has, at most, countably many ergodic components. (Hint: Imitate the Hopf argument perform in chapter 2. First of all notice that, by Fubini theorem, almost all segment has a full measure of point on which f^+ and f^- are well defined and equal. Thus almost all points have a stable manifold of some length with almost all point on it with unstable manifold of some length. This allows immediately to construct that almost all points belong to an invariant set of positive measure on which the map is ergodic. Clearly there can be at most countably many such sets.)
- 3.17** define cocycles (Hint: The derivative of the flow with collisions can be also naturally factored onto the quotient of the tangent bundle TM^c of M^c by the vector field F , which we denote by $\widehat{TM}^c$. Note that for a point $z \in \partial M^c$ the tangent to the boundary at z can be naturally identified with the quotient space. We will again denote the factor of the derivative cocycle by

$$A^t(x) : \widehat{T}_x M^c \rightarrow \widehat{T}_{\Psi_t^x} M^c.$$

We will call it the transversal derivative cocycle. If the derivative cocycle has well defined Lyapunov exponents then the transversal derivative cocycle has also well defined Lyapunov exponents which coincide with the former ones except that one zero Lyapunov exponent is skipped.)

Notes

This point of view (fixed point for manifolds) is brought to in extreme consequences in the pugh-shub.

Mentions conjugation with the linear part (Hartman-Grobman theorem)
Mentions Katok-Strelchyn